Smart Waste Segregation Assistant: Phase 2 Deep Transfer Learning, Hybrid Fusion, and Frozen Evaluation

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CSAI Project 12 — Phase 2: Deep Learning and Evaluation
13 November 2026

***Abstract*—This paper reports Phase 2 of a six-class waste material classifier on the TrashNet benchmark. Building on Phase 1's RBF SVM (test macro F1 0.8963), we evaluate deep transfer learning with MobileNetV2 and EfficientNet-B0, a systematic one-factor-at-a-time ablation study, and a late-fusion hybrid combining the fine-tuned CNN with the Phase 1 SVM. Temperature scaling is applied on training out-of-fold (OOF) predictions, and two abstention thresholds are pre-registered on the validation set before a single frozen test pass. The H3 late-fusion hybrid achieves macro F1 0.9233 [0.889, 0.952] and accuracy 93.63% on 377 held-out test images, significantly outperforming the A2.1 CNN ($p = 0.000019$, McNemar exact). The advantage over the Phase 1 SVM ($p = 0.035$) is suggestive but does not survive Bonferroni correction at $\alpha_{Bonf} = 0.0167$. An educational Gradio demo with per-class probabilities and an uncertainty banner uses frozen calibrated thresholds from `protocol.json`.**

***Index Terms*—Waste classification, MobileNetV2, EfficientNet, transfer learning, late fusion, temperature scaling, selective classification, McNemar test.**

I. Introduction

Automated visual waste sorting can support municipal recycling by routing items to the correct stream. Phase 1 established a reproducible evaluation protocol on the TrashNet benchmark [1]: a group-aware 70/15/15 split of 2,521 images, five frozen training folds, and an RBF SVM on 1,608 handcrafted color and texture features that reached test macro F1 0.8999. Phase 2 investigates whether deep transfer learning and hybrid fusion can improve on this baseline while respecting the same frozen test split and pre-registration discipline.

Three questions structure Phase 2: (1) Can a pretrained CNN adapted for TrashNet beat handcrafted features? (2) Does late probability fusion correct the residual errors of either modality? (3) Does temperature calibration on training OOF predictions transfer reliably to test? We answer each via a pre-registered protocol [2] in which all configuration decisions are made before the test split is touched.

II. Related Work

MobileNetV2 uses inverted residuals and linear bottlenecks to build an efficient deep feature extractor [3]. EfficientNet-B0 compounds depth, width, and resolution scaling for strong ImageNet performance with fewer parameters [4]. Yosinski et al. show that lower CNN layers learn generic features (edges, textures) transferable across domains, while upper layers are task-specific and benefit from fine-tuning [5]; this motivates our OFAT ablation across freezing depths.

Probabilistic calibration is especially important for selective classification [6]. Guo et al. show that modern neural networks are often overconfident and that temperature scaling is a simple, effective post-hoc fix [7]. We apply scalar temperature scaling [7] fitted on training OOF predictions to avoid any validation leakage.

Late fusion combines independent class-probability vectors from complementary models, reducing variance when their errors are largely disjoint [8]. Razavian et al. demonstrate that CNN features off-the-shelf outperform many purpose-built representations [9]; our work extends this by combining them with domain-engineered features rather than replacing them.

III. Data and Regularisation

(Data pipeline and regularisation: Om Yadav.)

We reuse the Phase 1 audit: 2,521 images partitioned group-aware into 1,764 training, 380 validation, and 377 test images (seed 42, manifest fingerprint `affd3e9d`). No new images are added in Phase 2 and the outer split is immutable.

For deep learning, images are resized to 224×224 with letterbox padding (white fill to preserve aspect ratio). Training augmentation applies random resized crop (scale 0.8–1.0), horizontal flip ($p = 0.5$), rotation ($\pm 10^\circ$), and color jitter (brightness/contrast 0.15, saturation 0.10), followed by ImageNet mean/std normalization. Validation and test images use center crop only. Balanced sampling and class-weighted loss are ablated as alternatives to standard training (see Sec. VI).

Augmentation is implemented as a togglable `build_transforms(train=True/False)` call that applies strictly to training loaders; validation and test use the identical deterministic pipeline. Five frozen training folds from Phase 1 are reused for all cross-validation, ensuring zero data leakage between selection and final evaluation.

IV. Architectures

(Backbone architectures: Aditya Bharadwaj.)

Two pretrained ImageNet backbones are evaluated:

- **MobileNetV2** [3]: 2.2M parameters, inverted residual blocks with depthwise separable convolutions, 1,280-dimensional penultimate embedding. We index its 19 structural layers (features 0–18); "last 2 blocks" refers to features 17–18 (the final InvertedResidual and the 1×1 pointwise conv).
- **EfficientNet-B0** [4]: compound-scaled variant with squeeze-and-excitation, 1,280-dimensional embedding.

In the frozen baseline (BASE), all backbone weights are frozen and only a 6-class linear head with $p = 0.2$ dropout is trained (7,686 parameters). In configuration `A2.1`, the last two MobileNetV2 blocks are unfrozen and trained at learning rate 10^{-4}, while the head uses 10^{-3}; total trainable parameters: 893,766.

Deterministic cached embeddings (SHA-256 provenance, stale-cache rejection) are pre-extracted for the H1 and H2 hybrid candidates evaluated in Sec. VII.

V. Training Procedure

(Training loop and scheduling: Shourya Bafna.)

Training uses AdamW [10] with decoupled weight decay 10^{-4}, cosine annealing with warm restarts [11] and $T_{\max} = 10$. Frozen backbone blocks are kept in `eval()` mode (BatchNorm and stochastic depth off) throughout training. Early stopping triggers when validation fold loss fails to improve for five consecutive epochs; maximum budget is 40 epochs.

The final A2.1 model is trained for exactly 13 epochs on the full 1,764 training images (the median best epoch from 5-fold CV), without touching validation data, and saved as `artifacts/phase2/final/A2.1/model.pt` (SHA-256 `26592f8b...5d28e`). Out-of-fold probability vectors are exported for downstream late fusion (H3) and temperature calibration.

VI. Ablation Study

A. Deep Learning Ablation Study

To isolate the empirical contributions of backbone architecture, fine-tuning depth, image resolution, head regularisation, data augmentation components, and class imbalance mitigation, we conducted a pre-registered one-factor-at-a-time (OFAT) ablation study comprising 13 configurations across the 5 frozen training cross-validation folds (65 evaluated folds, evaluated on Apple Silicon GPU hardware). The baseline model—a frozen MobileNetV2 [3] backbone with a linear classification head, $p = 0.2$ dropout, and comprehensive data augmentation at 224×224 resolution—achieved a mean Macro F1 of 0.8515 ± 0.0235 (87.19% accuracy). Under our pre-registered selection rule ($|\Delta| > \sigma = 0.0235$), swapping the frozen backbone to EfficientNet-B0 [4] (0.8557 ± 0.0263), adjusting dropout ($p = 0.0$: 0.8503, $p = 0.5$: 0.8479), omitting individual augmentations (e.g., removing $\pm 10^\circ$ rotation: 0.8490), removing all augmentation (0.8339 ± 0.0204, $\Delta = -0.0176$), and incorporating class weighting (0.8453) or balanced sampling (0.8419) yielded score differences within σ, exhibiting no clear difference under the pre-registered rule in overall Macro F1. However, class-weighted loss and balanced sampling explicitly improved mean per-fold trash recall from 65.63% to 72.89% and 72.84% respectively (pooled out-of-fold recall from 63/96 to 70/96). Conversely, reducing input resolution to 160×160 fell just beyond σ (worse, Macro F1 0.8271 ± 0.0257, $\Delta = -0.0244$). Crucially, partial fine-tuning of only the final two structural blocks in MobileNetV2 (`A2.1`, training 893,766 parameters while keeping the first 17 blocks frozen, with differential learning rates of 10^{-3} on the head and 10^{-4} on the backbone) produced a clear improvement under the rule to **0.8825 ± 0.0219 Macro F1** (89.80% accuracy, $\Delta = +0.0310 > \sigma$, median best epoch 13), driven by recall gains on trash (79.21% vs. 65.63%) and metal (94.39% vs. 89.50%). We note as a limitation that most frozen configurations reached their best epoch at 38–40 of the pre-registered 40-epoch cap (e.g., BASE median best 38, `A4.2` and `A5.1` median best 40) whereas `A2.1` early-stopped naturally (mean 20.2 epochs run); part of `A2.1`'s margin may reflect this budget constraint, and extended frozen training is left as future work. Configuration `A2.1` is on par with the Phase 1 SVM on handcrafted features (0.8803 ± 0.0215, $\Delta = 0.0022 \ll \sigma$), while Daksh's early feature fusion hybrid H2 (0.9044 ± 0.0332) shows no clear gain over `A2.1` under the σ rule ($\Delta = 0.0219 < 0.0332$), with the final evaluation deferred to upcoming late fusion (H3) and the held-out test set. Consequently, configuration `A2.1` was designated as the primary deep-learning model, and its out-of-fold predictions were exported for late fusion.

VII. Hybrid Models

A. Hybrid ML and Deep Learning Feature Fusion

While handcrafted photometric and texture descriptors achieved strong baseline classification in Phase 1 (0.8803 ± 0.0215 mean cross-validation macro F1 across five folds), systematic confusions persisted among glass, metal, and plastic due to overlapping specular reflections and smooth surface gradients. To investigate whether high-level geometric representations resolve these material confusions, we evaluated a hybrid architecture combining off-the-shelf deep convolutional neural network (CNN) representations [9] with engineered descriptors. Pretrained 1,280-dimensional feature embeddings from frozen MobileNetV2 and EfficientNet-B0 backbones were extracted and fused with Phase 1's winning 1,608-dimensional extended color and spatial texture features. To prevent the higher-dimensional handcrafted block from dominating the radial basis function (RBF) kernel distance metric, we enforced a fold-local dimensional normalization protocol: each block was independently standardized using training-fold statistics and scaled by $1/\sqrt{D_{\text{block}}}$ prior to concatenation ($D_{\text{total}} = 2,888$), giving both modalities equal total geometric variance. Evaluating across the five frozen training cross-validation folds, early feature fusion with an RBF SVM ($C = 10.0$, $\gamma =$ 'scale') achieved a mean cross-validation macro F1 of 0.9044 ± 0.0332 and an overall accuracy of 91.73% (compared to 0.8597 ± 0.0194 for classical classifiers trained on CNN embeddings alone).

In accordance with our pre-registered decision rule ($\Delta > \sigma$), the $+0.0241$ absolute gain of early fusion under initial discrete predictions (0.9044 ± 0.0332 Macro F1) did not constitute a clear win under the pre-registered rule as it remained within the cross-fold sample standard deviation ($\sigma = 0.0332$); when evaluated under the argmax of predicted probabilities for strict cross-model consistency, H2 achieves 0.9175 ± 0.0262 with 92.52% pooled accuracy (1,632/1,764). Following the selection of the fine-tuned MobileNetV2 architecture (A2.1, 0.8825 ± 0.0219 macro F1, 89.80% pooled accuracy) in Phase 2 ablations, we evaluated late probability fusion (H3) combining out-of-fold probability vectors from A2.1 and the calibrated Phase 1 SVM via convex blending ($p = w \cdot p_{\text{cnn}} + (1 - w) \cdot p_{\text{svm}}$). Analysis of out-of-fold predictions reveals that while CNN and SVM predictions disagree on 16.72% of training images (295/1,764), both models fail simultaneously on only 2.61% (46/1,764), demonstrating complementary error patterns. Using nested cross-validation across the five frozen training folds—where fusion weight $w \in [0, 1]$ was selected on inner training folds independently for each outer fold—equal weighting ($w = 0.5$) was selected across all folds. H3 achieved a mean cross-validation macro F1 of $\mathbf{0.9364 \pm 0.0171}$, mean per-fold accuracy of 94.62%, and pooled accuracy of $\mathbf{94.61}\%$ (1,669/1,764). Comparing against the best standalone single model (A2.1, 0.8825 macro F1), this represents a clear win under the pre-registered rule ($\Delta = 0.0539 > \sigma = 0.0219$). When compared to H2, $\Delta = 0.0189 < \sigma_{\text{H2}} = 0.0262$, so H3 is carried forward as the highest mean candidate per protocol rather than a clearly separated winner under the rule over H2. All cross-validation estimates remain somewhat optimistic as base models were selected on these five folds; the final test split ($N = 377$) provides the unbiased assessment. Pooled confusion analysis across all 1,764 training samples demonstrated that metal recall reached 98.25% (281/286), reducing metal-to-glass errors to just 3 (an 85.7% reduction compared to Phase 1), while plastic recall reached 91.67% (308/336) and trash recall rose to 86.46% (83/96). The final hybrid model was assembled on the full training split—preserving the verified A2.1 checkpoint (SHA-256: `26592f8b...5d28e`) without retraining—and exported for downstream probability calibration and operational abstention threshold selection on the held-out validation split.

VIII. Calibration and Final Evaluation

A. Frozen Phase 2 Test Evaluation and Statistical Comparison

Before accessing the untouched TrashNet test split ($N = 377$), all three finalists—Phase 1 RBF SVM, fine-tuned MobileNetV2 (`A2.1`), and equal-weight late fusion (`H3`, $w = 0.5$)—were frozen in `protocol.json` (git revision `62db0847`). Each model's scalar temperature T was fitted exclusively on its 1,764 training out-of-fold probabilities, and two abstention thresholds were selected strictly on the 380 validation images: a headline rule ($\geq 90\%$ accuracy, $\geq 80\%$ coverage) and a pre-declared demo rule ($\geq 95\%$ accuracy, $\geq 80\%$ coverage; added because `H3`'s headline threshold is 0.00). Model SHA-256 hashes were verified prior to the single test pass.

Table I reports the single test evaluation. `H3` Hybrid achieves the strongest test performance across every metric, reaching Macro F1 $\mathbf{0.9233}$ and accuracy $\mathbf{93.63}\%$ with the lowest expected calibration error (ECE $= 0.0130$, Brier 0.0971, log loss 0.2257). At its frozen demo threshold ($\tau = 0.53$), `H3` retains 96.55% coverage at 95.33% accepted accuracy, whereas `A2.1`'s strict demo threshold ($\tau = 0.87$) reaches 76.39% coverage on test, below its 80% validation target, showing that validation-chosen thresholds do not transfer exactly across splits. Compared against the historical Phase 1 test ensemble (0.8999 Macro F1, 90.19% accuracy, ECE 0.1325), the Phase 2 full-train `phase1_svm` matches accuracy exactly (90.19%, Macro F1 0.8963 vs 0.8999 because the refit SVM uses SVC's internal Platt scaling instead of Phase 1's calibrated ensemble) while OOF temperature scaling reduces test ECE by 72.8% to 0.0361.

Pairwise exact McNemar binomial tests on discordant unconditional predictions (Table II) confirm that `H3` Hybrid significantly outperforms `A2.1` CNN ($p = 0.00002 < 0.0167$, 23 vs. 2 discordant wins, surviving Bonferroni). Although `H3` leads Phase 1 SVM nominally (23 vs. 10, $p = 0.0351 < 0.05$), that advantage is suggestive but does not clear the strict Bonferroni threshold 0.0167 on $N = 377$ images (overlapping 95% bootstrap CIs [0.889, 0.952] vs [0.860, 0.926]); `A2.1` CNN and Phase 1 SVM do not differ significantly ($p = 0.3581$). Hook-based Grad-CAM on `A2.1`'s final convolutional block across 12 test images (6 correct, 6 errors) gives a qualitative assessment of spatial attention. Foreground is defined heuristically as pixels with HSV saturation > 0.12 or value < 0.92 (a colour heuristic, without object masks). While 11/12 heatmaps place $\geq 80\%$ of activation mass on the foreground object (mean 93.6%) rather than TrashNet's white posterboard, Grad-CAM does not explain why `A2.1` CNN trails Phase 1 SVM ($p = 0.358$, not significant), nor does it show complementary spatial features that lift `H3`; `H3`'s gain is demonstrated by McNemar discordant counts (23 vs 2 wins, $p = 0.000019$), not Grad-CAM. All six inspected errors involve clear object localization on overlapping glass, metal, and plastic shapes and specular reflections. External robustness remains unverified (robustness set not collected), leaving TrashNet background dependence, CV selection optimism, uncommitted working tree during H3 fitting (pinned in protocol.json), and low test support ($n = 20$ for trash) as primary limitations.

B. Grad-CAM Interpretability

Hook-based Grad-CAM [12] is applied to `A2.1`'s final convolutional block across a 12-image grid (6 correct spanning all classes, 6 high-confidence test errors spanning glass/metal/plastic confusions). The "foreground" heuristic is HSV saturation > 0.12 or value < 0.92 (a colour threshold, not an object mask); Table III reports the foreground activation share for each inspected case.

TABLE I
UNTOUCHED PHASE 2 TEST RESULTS ($N = 377$, 95% BOOTSTRAP CIS OVER 2,000 RESAMPLES, AND BOTH PRE-DECLARED FROZEN THRESHOLDS)

Model	T	Macro F1 [95% CI]	Accuracy [95% CI]	ECE	Headline (τ, Cov/Acc)	Demo (τ, Cov/Acc)
Phase 1 SVM	0.834	0.8963 [0.860, 0.926]	90.19% [87.0, 93.1]	0.0361	0.45 (97.6% / 91.6%)	0.76 (84.4% / 97.2%)
A2.1 CNN	1.103	0.8577 [0.813, 0.897]	88.06% [84.6, 91.2]	0.0367	0.50 (96.6% / 90.4%)	0.87 (76.4% / 95.5%)
H3 Hybrid	0.536	**0.9233** [0.889, 0.952]	**93.63%** [91.0, 96.0]	0.0130	0.00 (100.0% / 93.6%)	0.53 (96.6% / 95.3%)

TABLE II
PAIRWISE EXACT MCNEMAR TESTS ($\alpha_{\text{Bonf}} = 0.0167$)

Comparison	Both	A	B	Err	p	Sig.
H3 vs. A2.1	330	23	2	22	1.9×10^{-5}	Yes
H3 vs. SVM	330	23	10	14	0.0351	No
A2.1 vs. SVM	307	25	33	12	0.3581	No

TABLE III
A2.1 GRAD-CAM FOREGROUND ACTIVATION SHARE

Image	Outcome	True→Pred	FG CAM
`cardboard1`	Correct	card→card	98.3%
`glass10`	Correct	glass→glass	69.3%
`metal103`	Correct	metal→metal	98.7%
`paper106`	Correct	paper→paper	88.1%
`plastic1`	Correct	plastic→plastic	86.9%
`trash109`	Correct	trash→trash	98.0%
`glass307`	Wrong	glass→metal	93.0%
`glass102`	Wrong	glass→plastic	97.7%
`plastic233`	Wrong	plastic→glass	97.1%
`metal293`	Wrong	metal→glass	99.2%
`plastic467`	Wrong	plastic→metal	97.2%
`plastic398`	Wrong	plastic→glass	99.3%

Key finding: 11 of 12 heatmaps concentrate $\geq 80\%$ of activation mass on the foreground object (mean 93.6%), suggesting foreground-focused attention in these inspected examples, without ruling out background dependence. However, Grad-CAM *does not* explain the performance gap between A2.1 and Phase 1 SVM (McNemar $p = 0.358$, not significant); nor does it explain H3's gain, which is demonstrated by discordant-count analysis (23 vs 2, $p = 0.000019$). Shape-focused attention is insufficient for glass/metal/plastic disambiguation because those materials share overlapping silhouette geometry.

IX. DEMO

The Gradio 6.x application loads all three Phase 2 models once per process from frozen artifacts and reads temperatures and demo thresholds from `protocol.json`. A model selector (H3 hybrid recommended, A2.1 CNN, Phase 1 SVM) lets users compare all three in a single interface. Predictions below the frozen demo threshold trigger an amber "Uncertain — manual inspection recommended" banner. SHA-256 hash verification at startup rejects mismatched or missing model files. The demo is for educational purposes; local recycling rules determine actual disposal.

X. LIMITATIONS

1) **Cross-validation optimism:** All models reuse the same 5 training folds for model selection; reported CV scores are optimistic ($H3$ CV 0.9364 vs test 0.9233; A2.1 CV 0.8825 vs test 0.8577).
2) **Small test set ($N = 377$, trash $n = 20$):** With only 33 discordant predictions between H3 and SVM, the nominal $p = 0.035$ does not survive Bonferroni correction ($\alpha_{Bonf} = 0.0167$).
3) **TrashNet white background:** All images were photographed against a uniform white posterboard; the robustness set was not collected (`docs/robustness/` absent on `main`).
4) **OOF temperature applied to full-train models:** Temperature was fitted on fold-model OOF predictions but applied to the full-training final models, following Phase 1 practice.
5) **H3 from uncommitted working tree:** `protocol.json` records `working_tree_dirty` in H3's metadata; model and OOF SHA-256 hashes are pinned and verified before every test prediction.
6) **Threshold fragility:** A2.1's demo threshold (0.87) achieved 82.4% validation coverage but only 76.4% test coverage, showing that tight confidence thresholds do not transfer exactly across splits.

XI. CONCLUSION

Phase 2 shows that partial fine-tuning of the last two MobileNetV2 blocks matches the Phase 1 handcrafted SVM (A2.1 CV 0.8825 vs SVM 0.8803) and that late probability fusion of complementary error patterns yields a clear improvement (H3 test macro F1 **0.9233**, significantly better than A2.1, $p = 0.000019$, Bonferroni-corrected). Temperature calibration on OOF data reduces ECE from 0.095 to 0.019 for H3. The frozen evaluation protocol with pre-registered thresholds ensures an auditable, single-pass test result.

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TABLE IV
PHASE 2 TECHNICAL CONTRIBUTIONS

Member	Primary contribution
Om Yadav	Phase 2 data pipeline, augmentation, balanced sampling
Aditya Bharadwaj	Backbone loaders, OFAT ablation study (13 configs, 65 folds)
Shourya Bafna	Training loop, AdamW/cosine schedule, early stopping
Daksh Batra	H1/H2 hybrids, H3 late fusion, error complementarity analysis
Prajjwal Tripathi	Temperature scaling, abstention thresholds, frozen test eval
Saksham Miglani	Demo, Phase 2 report and presentation assembly

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